

Problem B. Bread

Time Limit: 1 seconds | Memory Limit: 512 MB

Standard Input / Standard Output



Natsu is eating bread at Sensei's café when a piece of bread falls onto the floor and starts rolling.

The fallen bread can be regarded as a convex polygon in a two-dimensional plane. Its n vertices, listed in counterclockwise order, are $A_1, A_2, \dots, A_n$. The vertices are pairwise distinct, and no three vertices are collinear. The floor can be regarded as the horizontal line $y = 0$.

Initially, the bread touches the floor along edge (A_n, A_1) . Let the position of vertex A_n at this moment be the origin. The bread rolls in the positive direction of the x -axis; that is, $A_n = (0, 0)$ and $A_1 = (x_1, 0)$, where $x_1 > 0$. The bread then starts rolling. The rolling process can be abstracted as an infinite sequence of rotations performed in order. In the i -th rotation, the entire bread is rotated clockwise about the current position of vertex $A_{((i-1) \bmod n)+1}$ until $A_{(i \bmod n)+1}$ touches the floor for the first time, that is, until its y -coordinate becomes 0 for the first time.

As a romantic philosopher, Natsu considers the rolling of the bread to be a coincidental and romantic event. For the sake of this romance, after the k -th rotation, Natsu records the distance $L(k)$ rolled by the bread and the area $S(k)$ swept out by the bread. You need to calculate their ratio as k tends to infinity, namely:

$$\lim_{k \rightarrow +\infty} \frac{S(k)}{L(k)}$$

The distance $L(k)$ rolled by the bread after the k -th rotation is defined as the distance from the origin to the position of vertex $A_{(k \bmod n)+1}$ when the k -th rotation ends. The area $S(k)$ swept out by the bread after the k -th rotation is defined as the area of the union of all regions covered by the convex-polygon bread at all moments from the beginning of the first rotation to the end of the k -th rotation.

Input

The first line contains one positive integer T ($1 \leq T \leq 10^3$), denoting the number of test cases.

For each test case, the first line contains one integer n ($3 \leq n \leq 10^5$), denoting the number of vertices of the convex-polygon bread.

The next n lines each contain two integers x_i, y_i ($0 \leq |x_i|, y_i \leq 10^9$). The i -th line denotes the coordinates of vertex A_i when the bread has just fallen onto the floor.

It is guaranteed that y_1, x_n , and y_n are all 0, and that $x_1 > 0$.

It is guaranteed that vertices $A_1, A_2, \dots, A_n$ are pairwise distinct, no three vertices are collinear, and they form a convex polygon in counterclockwise order.

It is guaranteed that the sum of n over all test cases in one test file does not exceed 3×10^5 .

Output

For each test case, output one floating-point number on one line, denoting the answer. Your answer is considered correct if and only if its relative or absolute error compared with the standard answer does not exceed 10^{-9} .

Suppose that your answer is a and the standard answer is b . Your answer is considered correct if

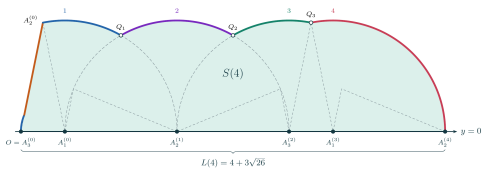
$$\frac{|a - b|}{\max\{b, 1\}} \leq 10^{-9}$$

Examples

Sample Input 1	Sample Output 1
2	4.908664519941197
3	1.384172075579563
2 0	
1 5	
0 0	
4	
1 0	
1 1	
0 1	
0 0	

Note

For the first test case, the process from the beginning of the first rotation to the end of the fourth rotation is shown below:



When $k = 4$, we have:

$$L(k) = 4 + 3\sqrt{26} \approx 19.297058541$$
$$S(k) = \frac{120}{13} + 13\sqrt{3} + \frac{91\pi}{6} + 43 \arctan \frac{1}{5} \approx 87.882927382$$